



QUALITY ASSURANCE



Quality Assurance

Quality assurance is typically focused on adherence to proper processes, in order to provide confidence that the appropriate levels of quality will be achieved.



Manual Testing



Objectives

- Preventing as many defects and critical bugs as possible by careful examination.
- To verify the fulfillment of all specified requirements.
- To build confidence in the level of quality of the test object.
- To provide sufficient information to stakeholders/business owner to allow them to make informed decisions, especially regarding the level of quality of the test object
- To find defects and failures thus reduce the level of risk of inadequate software quality.



Errors, Defects, and Failures

Error: A mistake, misconception, or misunderstanding on the part of a software developer.

Defect: A deviation from expected software behavior caused by error

Failure: Inability of a software system or component to perform as required. Result of a defect



Test Types

A test type is a group of test activities aimed at testing specific characteristics of a software system, or a part of a system, based on specific test objectives



White-box Testing

- White-box testing derives tests based on the system's internal structure or implementation
- Internal structure may include code, architecture, work flows, and/or data flows within the system
- White-box test design and execution may involve knowledge, such as the way the code is built, how data is stored

Some types of white-box tests

- Unit Testing
- Testing for Memory Leaks
- Penetration Testing
- Path testing



Black-box Testing

- In Black-box testing, a tester doesn't have any information about the internal working of the software system
- Black box testing is a high level of testing that focuses on the behavior of the software
- Black box testing can be applied to virtually every level of software testing: unit, integration, system, and acceptance.

Some types of black-box tests

- Functional Testing
- Non-functional testing
- Regression Testing



Functional Testing

- Functional testing of a system involves tests that evaluate functions that the system should perform.
- Functional requirements may be described in work products
- Functional tests should be performed at all test levels
- Functional testing considers the behavior of the software

Some types of functional tests

- | | |
|----------------------|---------------------|
| • Unit testing | Component testing |
| • Smoke testing | Sanity testing |
| • Regression testing | Integration testing |
| • API testing | UI testing |



Non-functional Testing

- Non-functional testing of a system evaluates characteristics of systems and software such as usability, performance efficiency or security
- Non-functional testing is the testing of “how well” the system behaves
- Non-functional testing can and often should be performed at all test levels, and done as early as possible
- Black-box techniques may be used to derive test conditions and test cases for non-functional testing

Some types of non-functional tests

- | | |
|---------------------|------------------------------|
| • Performance Tests | Load Tests |
| • Stress Tests | Volume Tests |
| • Security Tests | Upgrade & Installation Tests |
| • Recovery Tests | |



Change-related Testing

- It is designed to verify that a change to the system has been fixed correctly and does not have any adverse effects on the system as a consequence
- After a defect is fixed, the software need to be tested with all test cases that failed due to the defect
- It is possible that a change made in one part of the code, whether a fix or another type of change, may accidentally affect the behavior of other parts of the code

Some types of change-related testing

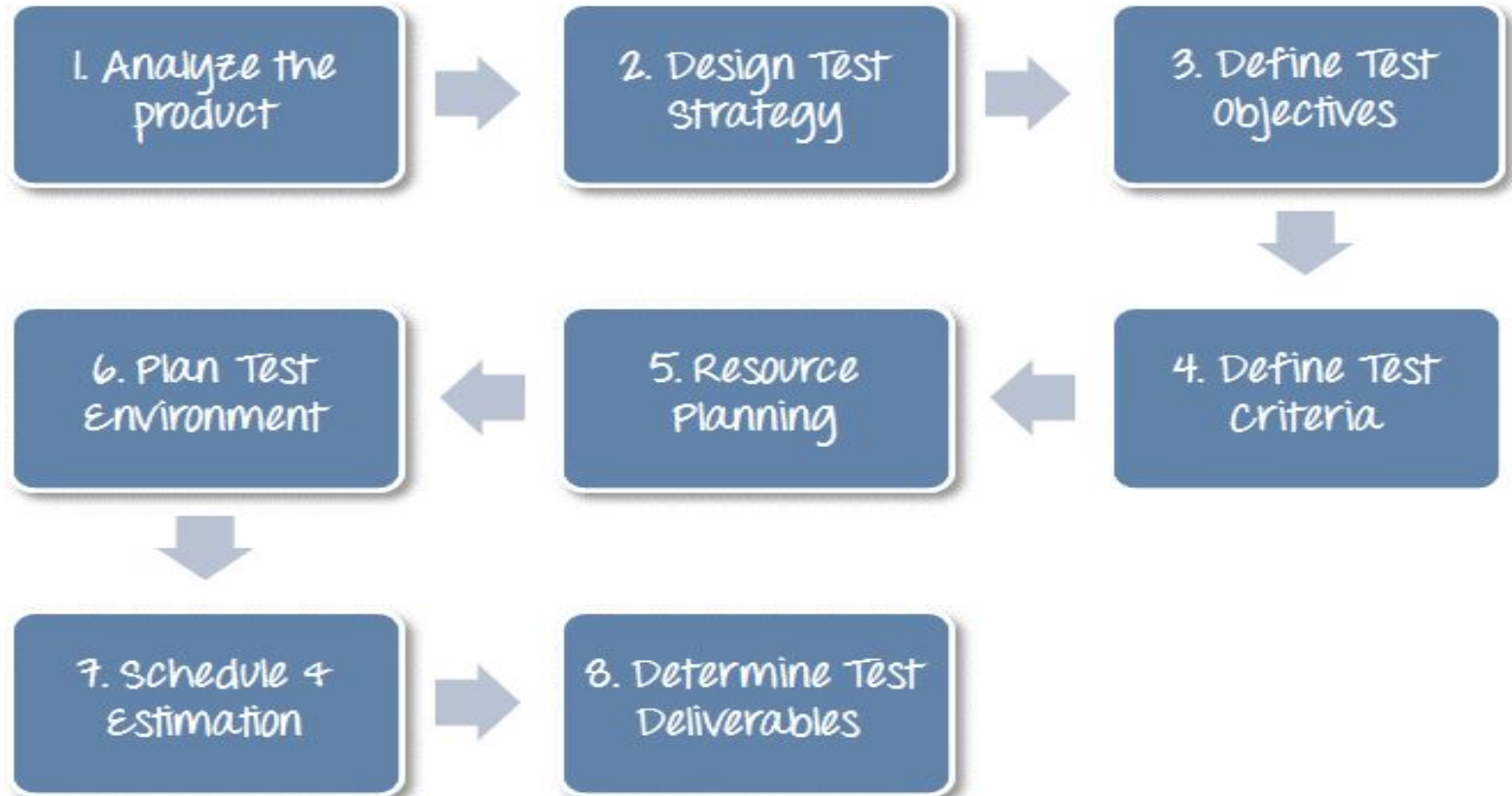
- Functional Testing
- Smoke Testing
- Regression Testing
- Build Verification Test
- Sanity Testing



Test Plan

- A test plan is a document detailing the objectives, resources, and processes for a specific test for a software or hardware product.
- The plan typically contains a detailed understanding of the eventual workflow.
- A document describing the scope, approach, resources, and schedule of intended test activities.

What should be included in a test plan?





Test Data

- Input given to a software program during test execution.
- It represents data that affects or affected by software execution while testing.
- Test data is used for both positive testing to verify that functions produce expected results for given inputs and for negative testing to test software ability to handle unusual, exceptional or unexpected inputs.



Test Case

- A Test Case is a set of actions executed to verify a particular feature or functionality of your software application.
- A Test Case contains test steps, test data, precondition, postcondition developed for specific test scenario to verify any requirement.



Requirements Traceability Matrix (RTM)

- Requirement Traceability Matrix (RTM) is a document that maps and traces user requirement with test cases.
- It captures all requirements proposed by the client and requirement traceability in a single document, delivered at the conclusion of the Software development life cycle.
- It validates that all requirements are checked via test cases such that no functionality is unchecked during Software testing.



Test Management

- The method consists of organizing, controlling, ensuring traceability and visibility of the testing process in order to deliver the high quality software application.
- Test Management is not just a single activity. It consists of a series of activities.



Test Organization

- Test Organization in Software Testing is a procedure of defining roles in the testing process.
- It defines who is responsible for which activities in testing process.
- Test functions, facilities and activities are also explained in the same process.
- The competencies and knowledge of the people involved are also defined however everyone is responsible for quality of testing process.



Test Estimation(Planning)

- Test Estimation is a management activity which approximates how long a Task would take to complete.
- Basically determination of how long and how much the testing cost



Resources



Times



Human Skill



Cost

Estimate of what?



Test Monitoring and Control

- Test Monitoring is a process in which the testing activities and testing efforts are evaluated in order to track current progress of testing activity, finding and tracking test metrics, estimating the future actions based on the test metrics and providing feedback to the concerned team as well as stakeholders about current testing process.
- Test Control in test execution is a process of taking actions based on results of the test monitoring process.



How to monitor and control test?

1. Create monitoring plan
2. Update progress record
3. Analyze record and make adjustments
4. Procedure the report



Risks Analysis

- Risk is the probability of occurrence of an undesirable event.
- Risk should be identified and corresponding solutions should be determined before the start of the project.



How to perform risk analysis?

1. Identify the Risks
2. Analyze Impact of each Identified Risk
3. Take countermeasures for the identified & Analyzed risk



Defect Management

- Defect Management is a systematic process to identify and fix bugs.



Defect Management Process

1. Discovery of Defect
2. Defect Categorization
3. Fixing of Defect by developers
4. Verification by Testers
5. Defect Closure
6. Defect Reports